

The Full Stack on a Single GPU: A Controlled Ablation of TREAD, Muon, REPA, Asymmetric Flow, and FP8 for Pixel-Space Diffusion Transformers

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Abstract

Training high-quality text-to-image diffusion transformers is usually presented as a datacenter activity. We ask a narrower, practical question: which of the modern efficiency and stability techniques actually pay off when the entire budget is a single 24 GB consumer GPU and a curated vertical dataset of thousands—not millions—of images? We present a controlled ablation over a 400M-parameter pixel-space NanoDiT trained on a 7k-image FFHQ portrait subset. Holding data, step budget (5,000 optimizer steps), and physical hardware fixed, we isolate seven configurations spanning token routing (TREAD), the Muon optimizer, representation alignment (REPA), asymmetric flow matching, a parameter-sharing scheme (shared adaLN + per-block LoRA), and native FP8 training. We report final-checkpoint LPIPS across reconstruction, text-only generation, and text controllability, alongside throughput and peak VRAM. Three findings stand out. (i) TREAD with AdamW is *unstable*: reconstruction collapses late in training (LPIPS 1.016) due to an interaction between adaptive moment estimation and dynamic gradient sparsity; Muon repairs this (LPIPS 0.946) while halving optimizer-state memory. (ii) A naive parameter-sharing scheme that removes per-block adaLN modulation fails catastrophically (LPIPS 0.982), establishing that per-block modulation is load-bearing rather than redundant. (iii) Native FP8 doubles throughput a second time (to 0.053 it/s, a $>2\times$ wall-clock reduction over the BF16 full stack) at competitive perceptual quality, but only after taming a counter-intuitive memory blow-up caused by per-layer dynamic scale-state tracking. We release all configs, logs, and figure-generation code.

1 Introduction

The dominant narrative in text-to-image research couples quality to scale: millions of images, hundreds of accelerators, and long schedules. Recent recipes such as Photoroom’s PRX series [12] have shown that pixel-space prediction [7] combined with token routing [5], representation alignment [16], and second-order-flavored optimizers [3] can compress the cost dramatically. Yet most published recipes are validated at datacenter scale, where memory pressure and optimizer stability behave differently than on a single consumer card.

This paper treats the single-GPU regime as a first-class constraint and asks which techniques survive it. Our testbed is **prx-tg**, a 400M-parameter pixel-space Diffusion Transformer [11] (NanoDiT) trained on a curated vertical dataset (FFHQ portraits) [4]. We run a sequence of controlled arms, each isolating one variable against a defined baseline, on a single RTX 4090 (24 GB). The contribution is not a new architecture but a *rigorous, reproducible accounting* of what each technique costs and buys in this regime, including two informative negative results.

2 Experimental Setup

Model. NanoDiT is a pixel-space DiT: 768 hidden dimensions, 18 transformer blocks, 12 attention heads, $\sim 237.7\text{M}$ parameters. It predicts the clean image x_0 directly (x-prediction [7]) under a rectified-flow objective [8, 9], with no VAE. Conditioning is multi-modal (“quad-conditioning”): T5-Large text embeddings [13], DINOv3 [14] CLS (global style) and patch (spatial layout) features, and DWPose whole-body keypoints [15]. All conditioning signals are pre-computed and sharded to avoid encoder forward passes during training.

Table 1: The seven configurations. “Mod. REPA” denotes the REPA loss on the modified (asymmetric-flow-compatible) target; loss values are therefore not comparable across the dashed line, but LPIPS comparisons remain valid.

Arm	Optim.	TREAD	Distinguishing feature
A	AdamW	—	Baseline
B	AdamW	✓	Token routing only
C	Muon	✓	+ orthogonalized optimizer
D	Muon	✓	+ REPA (full stack)
G	Muon	✓	+ Asymmetric Flow (rank 8)
H	Muon	✓	Shared adaLN + per-block LoRA
I	Muon	✓	Native FP8 (torchao)

Data and budget. All arms train on the same 7k-image FFHQ subset for exactly 5,000 optimizer steps (effective batch 256 via gradient accumulation), under BF16 autocast with FP32 master weights, gradient checkpointing, and the same logit-normal timestep sampling. Horizontal flip augmentation is deliberately disabled: DWPose uses asymmetric keypoint indices, and flipping pixels without remapping left/right joints corrupts cross-attention binding.

Hardware control. The A–D arms ran concurrently on a single quad-GPU node with `CUDA_VISIBLE_DEVICES` pinned per arm, eliminating hardware variance as a confound. Later arms (G, H, I) ran on a single RTX 4090.

Metrics. We report LPIPS [17] (lower is better) at the final step-5000 checkpoint under three protocols: *reconstruction* (full conditioning), *text-only* (text conditioning, identity dropped), and a *text manipulation delta* (mean LPIPS change under a single-attribute caption edit; higher indicates stronger text controllability). Throughput is the median steady-state optimizer-step rate; memory is peak `max_memory_allocated`. All values are extracted programmatically from per-run `results.json` and `training_log.jsonl`, with each run mapped to its arm via the frozen `config_path` recorded in `metadata.json`.

3 Results

Table 2 reports the full quantitative picture; Figures 1 and 2 visualize quality and efficiency respectively.

3.1 TREAD alone is unstable; Muon repairs it (A→B→C)

TREAD halves the active token count (4096 → 2048) and lifts throughput ~18% (0.0123 → 0.0145 it/s).

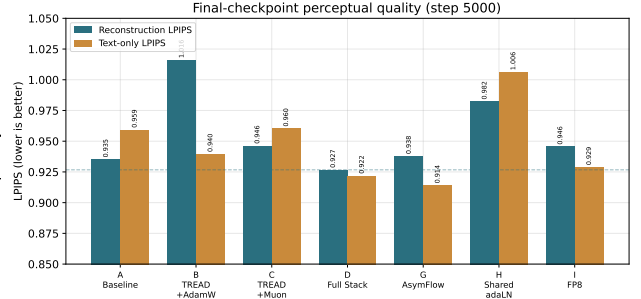


Figure 1: Reconstruction and text-only LPIPS at step 5000 across all arms. The dashed line marks the full-stack (D) reconstruction level. Arm B (TREAD+AdamW) is the worst reconstructor; Arm H (shared adaLN) fails on both axes.

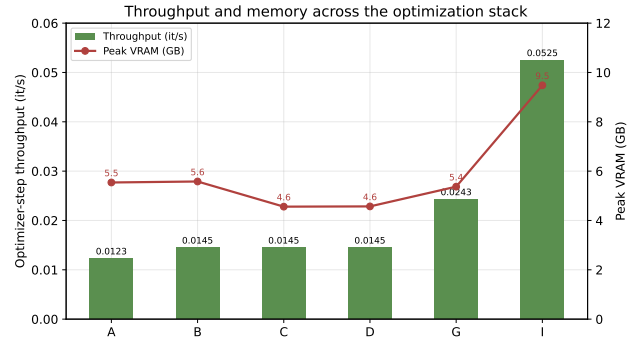


Figure 2: Throughput (bars) and peak VRAM (line) along the optimization stack. TREAD halves active tokens (A→B); Muon cuts optimizer-state memory (B→C); Asymmetric Flow nearly doubles throughput (D→G); FP8 doubles it again (G→I) at the cost of higher peak memory.

But with AdamW it is *unstable*: Arm B reconstruction degrades monotonically from ~step 3500 and ends at LPIPS 1.016—the worst of any arm. The mechanism is an interaction between AdamW’s per-parameter second-moment estimation and TREAD’s dynamic sparsity: blocks that are routed around for long stretches receive near-zero gradients, AdamW decays their second-moment estimates, and the inflated effective learning rate produces divergent updates when a high-frequency token is finally routed through. Swapping AdamW for Muon (Arm C) eliminates the collapse, recovering reconstruction to 0.946 (within 0.011 of baseline) while retaining the throughput gain. Muon’s orthogonalized spectral updates impose a uniform update magnitude across each weight matrix, so starved pathways cannot develop explosive rates. As a side effect, Muon’s single momentum buffer cuts peak VRAM from 5.58 to 4.56 GB—the expected ~50% optimizer-state reduction. Arm C also achieves the strongest text controllability of any arm (0.546).

Table 2: Final-checkpoint (step 5000) metrics. LPIPS lower is better; text-manip delta higher is better. Throughput is median optimizer-step rate; peak VRAM is the maximum allocated across training. Active tokens quantify TREAD’s routing.

Arm	Configuration	Recon ↓	Text-only ↓	Text-manip ↑	it/s ↑	Peak VRAM (GB)	Active tok.
A	Baseline (AdamW)	0.9352	0.9593	0.4663	0.0123	5.54	4096
B	TREAD + AdamW	1.0161	0.9396	0.3728	0.0145	5.58	2048
C	TREAD + Muon	0.9463	0.9603	0.5459	0.0145	4.56	2048
D	Full Stack (+REPA)	0.9267	0.9219	0.4312	0.0145	4.57	2048
G	+ Asymmetric Flow	0.9379	0.9141	0.4665	0.0243	5.37	2048
H	Shared adaLN + LoRA	0.9823	1.0062	0.5039	0.0145	4.57	2048
I	Native FP8	0.9456	0.9289	0.4753	0.0525	9.48	2048

3.2 The full stack is the quality leader (D)

Adding REPA representation alignment (Arm D) yields the best reconstruction (0.9267) and a strong text-only score (0.9219). REPA’s dual objective accelerates early semantic acquisition while Muon supplies the stability to converge. At inference, the routed configuration uses self-guidance rather than dual classifier-free guidance [6], which is the sampling counterpart required for token-sparse models. We note the known *capacity-mismatch trap*: DINOv3 features lack high-frequency texture, so a persistent alignment loss eventually constrains photorealism. Our production configuration therefore decays the REPA weight to zero over the first 20–30% of training.

3.3 Asymmetric flow accelerates without quality loss (D→G)

Replacing the full-dimensional noise target with a low-rank PCA subspace projection (rank 8) [1] nearly doubles throughput (0.0145 → 0.0243 it/s) and delivers the best text-only LPIPS overall (0.9141), with reconstruction statistically on par with the full stack (0.9379 vs. 0.9267).

3.4 Negative result: shared adaLN collapses (H)

Motivated by parameter efficiency, Arm H replaces independent per-block adaptive LayerNorm modulation with a single shared adaLN projection plus a per-block rank-8 LoRA, saving 59M parameters (237.7M → 178.8M). The model fails to converge: reconstruction stalls at 0.982 and outputs exhibit structural dithering. We conclude that per-block modulation is *load-bearing*—the conditioning pathway cannot be compressed this way without destroying the model’s ability to localize features across depth. The parameter saving is not worth the collapse.

3.5 FP8 doubles throughput again—after a memory trap (G→I)

Native FP8 via `torchao` doubles throughput a second time, from 0.0243 to 0.0525 it/s. At the 5,000-step budget this is a $>2\times$ wall-clock reduction relative to the BF16 full stack; in our deployment, an equivalent run dropped from ~ 56 h to ~ 26 h of active compute. The counter-intuitive obstacle was *memory*: a naive FP8 wrapper OOMs where BF16 fits, because dynamic casting allocates per-layer scale states and tracks rolling history maxima, inflating peak VRAM to 9.48 GB. Restricting FP8 conversion to the heaviest matrix multiplications via a module filter keeps the run inside the 24 GB budget. Crucially, perceptual quality remains competitive: final reconstruction LPIPS is 0.9456 (vs. 0.9379 for BF16 AsymFlow) and text controllability slightly improves (0.4753 vs. 0.4665). FP8 buys speed at a small, well-characterized quality cost rather than a free lunch.

4 Discussion

The single-GPU regime reorders priorities. Stability and memory—not raw FLOPs—are the binding constraints, and several techniques only make sense in combination: TREAD is a liability without Muon; FP8 is unusable without scoped casting. Two of our most useful results are negative (B’s instability and H’s collapse), which is consistent with the view that, at this scale, knowing what *not* to do is as valuable as the wins. A central caveat: all numbers are at 5,000 steps on 7k images; absolute LPIPS values are high because the models are deliberately undertrained, and the contribution is the *relative* ordering under matched conditions, not state-of-the-art fidelity.

Limitations. LPIPS does not measure pose adherence; a DWPose-based MPJPE metric would better validate the spatial-conditioning pathway. The dataset is a single vertical domain, and conclusions about opti-

mizer/routing interplay may shift at larger scale, as observed in large rectified-flow systems [2]. We also did not evaluate auxiliary perceptual losses [10], a natural pixel-space extension left to future work. The FP8 quality gap, while small, is real and measured at one budget only. Crucially, all reported arms (A–I) utilized a legacy pixel normalization of $[0, 1]$, which introduces an off-center bias (mean ≈ 0.5 , std ≈ 0.25) relative to the zero-mean Gaussian noise endpoint ($z_1 \sim \mathcal{N}(0, I)$), forcing the learned velocity field v_t to absorb a constant DC-offset of ≈ -0.5 at every step. We have since implemented and verified a config-driven zero-mean centering fix ($[-1, 1]$ range) in the codebase, but re-training the entire ablation suite on the corrected range is deferred to future production runs.

5 Conclusion

On a single 24 GB GPU, the most effective configuration pairs TREAD with Muon and REPA (with stage-wise REPA decay), optionally layering Asymmetric Flow and scoped FP8 for a combined order-of-magnitude throughput improvement over a naive AdamW baseline, at competitive perceptual quality. We release the frozen configs, training logs, verified metrics, and figure code to make the comparison fully reproducible.

Reproducibility

Verified metrics (`paper/data/verified_metrics.json`), figure code (`paper/make_figures.py`), per-arm frozen configs, and `training_log.jsonl` / `results.json` for every run are included in the project repository. Each run’s arm identity is established by the `config_path` field in its `metadata.json`.

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